The value of a CAP on frozen GPT-2 small: frozen vs BP-CAP vs PC-CAP

Capstan (pc_cap)

2026-10-20

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Supplement to study `ext-20261009` · Capstan · 2026-10-20 · pc_cap code revision `94f5917deb29` · requested by the lead on 2026-10-10 ("Frozen/no-CAP baseline must be central"). Built from the existing recorded scores of `ext-20261009` (no new GPU evaluations); the frozen record was not modified.

Two-part question. A. What is gained by adding a CAP (episodic memory + learned reader, 3,348,228 parameters, writes at blocks 4/8/12) to frozen GPT-2 small, especially on extreme examples? **B.** Given that a CAP is beneficial, does training its reader by error predictive coding (PC-CAP) offer any advantage over backprop (BP-CAP)? Question A is primary here; question B is reported inside it.

Models. *Frozen GPT-2 small (no CAP)*: the base parameters (digest `c4ac3fb8…`) loaded directly; validated against the record's cap-off path (direct forward reproduces saved cap-off losses to 5e-5 nats; a reader that abstains returns the base logits exactly). *BP-CAP* and *PC-CAP*: the frozen record's PC-reader study — the same cap architecture, the same adjoint acquisition (5 delta steps), the same 300 edits of realization 0 / order 100, three paired training seeds each; only the reader's training-time gradient estimator differs. Cap numbers are shown as the **seed mean** (per-probe mean over the three seeds) with the per-seed values beside them; seeds are not independent populations.

Conventions. Principal horizon: the 300-edit memory (100-edit rows are in the CSVs). Losses: teacher-forced per-token NLL (nats/token) of the stated target on identical serialised prompt/answer pairs for every model. Exact match: greedy, ≤ 32 tokens, newline stop, normalised alias match (the project's registered convention). Intervals: 95 % percentile bootstrap over items (or endpoint rows), the same resampled indices for every model (pairing preserved); 2,000 draws, 500 for deciles. "D" = L_cap − L_frozen on the same probe; a probe is "unchanged" when $|D| \leq$ 1e-3 nats (float32 reduction noise is ~1e-4).

1. Executive summary

- **A1. Ordinary adaptation.** On paraphrases of the taught facts (the cap never saw these prompts), mean per-token NLL falls from frozen to cap: zsRE 6.10 → 0.13 (BP-CAP, −98 %) / 0.20 (PC-CAP, −97 %); CounterFact 7.37 → 1.31 (BP-CAP, −82 %) / 2.43 (PC-CAP, −67 %); MQuAKE 6.05 → 1.59 (BP-CAP, −74 %) / 2.09 (PC-CAP, −65 %). Exact-match paraphrase retention rises from a measured 0 % (frozen) to zsRE 98.0 % (BP) / 96.6 % (PC); CounterFact 81.5 % (BP) / 66.9 % (PC); MQuAKE 67.4 % (BP) / 59.6 % (PC). On the taught prompts themselves both caps reach 99–100 % (frozen: 0 % by the eligibility screen).

- **A2. The frozen model's hardest examples.** On the fixed 5 % of paraphrases that frozen GPT-2 finds hardest, mean NLL falls zsRE (k = 15): 11.37 → 0.05 (BP, −100 % vs −98 % overall) / 0.05 (PC, −100 % vs −97 % overall); exact match on the set 100.0 % / 100.0 %; CounterFact (k = 30): 10.42 → 2.05 (BP, −80 % vs −82 % overall) / 2.78 (PC, −73 % vs −67 % overall); exact match on the set 78.9 % / 74.4 %; MQuAKE (k = 15): 10.17 → 3.13 (BP, −69 % vs −74 % overall) / 3.13 (PC, −69 % vs −65 % overall); exact match on the set 33.3 % / 33.3 %. The absolute reduction is largest where the frozen model is worst (decile figures), while the caps' own loss also rises with frozen difficulty, so the relative reduction on the hardest cases is somewhat smaller than overall on CounterFact and MQuAKE and complete on zsRE.

- **A3. Each model's own worst errors.** The caps' own worst 5 % and 1 % of paraphrase losses are lower than the frozen model's own worst cases, but by much less than the mean improvement: zsRE: CVaR95 11.33 → 2.05 (BP, −82 %) / 3.36 (PC, −70 %); CVaR99 12.16 → 5.27 / 5.75; CounterFact: CVaR95 10.40 → 7.93 (BP, −24 %) / 8.78 (PC, −15 %); CVaR99 11.27 → 9.21 / 9.55; MQuAKE: CVaR95 10.12 → 6.84 (BP, −32 %) / 7.62 (PC, −25 %); CVaR99 11.37 → 8.82 / 8.92. A cap's residual worst cases are the prompts on which its reader abstained (loss = frozen loss to the bit) or fired without fixing the answer (table 9.2).

- **A4. Collateral change.** Against frozen GPT-2 directly (not only against each cap's own cap-off reference), the caps leave most locality-type prompts bit-identical; the exceptions are rare but can be severe: zsRE: worse than frozen on 0.0 % (BP) / 0.0 % (PC) of item-locality prompts, severe (> 2 nats/token) events 0 / 0 over three seeds (296 prompts each); CounterFact: worse than frozen on 0.0 % (BP) / 0.3 % (PC) of item-locality prompts, severe (> 2 nats/token) events 0 / 2 over three seeds (900 prompts each); MQuAKE: worse than frozen on 1.1 % (BP) / 1.1 % (PC) of item-locality prompts, severe (> 2 nats/token) events 4 / 8 over three seeds (560 prompts each). On 245,237 positions of unrelated text per cell the mean change is 0.0e+00 to 6.3e-03 nats, with single-position maxima of 0.0–13.1 nats and 0–419 positions changed by more than 1 nat per cell (table 8.3).

- **A5. What the CAP does not fix.** MQuAKE multi-hop composition stays at the frozen level: all-three-questions success 0.0 % (BP) / 0.0 % (PC) of 80 cases, question accuracy 0.6 % / 0.4 % vs frozen 0.0 %; and every abstained paraphrase keeps the frozen model's full loss.
- **A6. Tail shape.** At common absolute thresholds (frozen P95 and 8 nats) 42 GPD fits with ≥ 50 exceedances have shapes from -0.42 to 0.32; 3 interval(s) lie entirely above zero (BP-CAP seed 0 on MQuAKE at frozen P95 ($\kappa = 0.30$ [0.04, 0.51], 75 exceedances); PC-CAP seed 0 on MQuAKE at frozen P95 ($\kappa = 0.32$ [0.01, 0.66], 67 exceedances); PC-CAP seed 2 on MQuAKE at frozen P95 ($\kappa = 0.32$ [0.04, 0.61], 68 exceedances)), and that cell turns negative at 8 nats. At every common threshold both caps have fewer exceedances than frozen GPT-2 (table 7.1): the caps thin the frozen tail rather than change its class, and none of these finite-sample fits is evidence of a power law.
- **B. PC vs BP within the CAP benefit.** The paired seed-mean contrast PC − BP on paraphrase NLL (positive favours PC) is zsRE -0.076 [-0.136, -0.027]; CounterFact -1.117 [-1.313, -0.916]; MQuAKE -0.495 [-0.651, -0.347] nats/token: a small fraction of either cap's gain over frozen, and negative on every dataset. PC-CAP retains fewer paraphrases than BP-CAP on CounterFact and MQuAKE in all three seeds and is close on zsRE. Its gate is more conservative, which shows up as fewer false fires and less unrelated-text harm on zsRE (zsRE false fires 9.7 % (BP) vs 3.7 % (PC), text-harm ES99+ 0.0143 vs 0.0016; CounterFact false fires 0.3 % (BP) vs 0.3 % (PC), text-harm ES99+ 0.2930 vs 0.3266; MQuAKE false fires 0.0 % (BP) vs 0.0 % (PC), text-harm ES99+ 0.2302 vs 0.2322), not as better behaviour on hard cases: in no extreme-case view (frozen-hardest sets, own worst cases, deciles, severe collateral events) is the PC-trained reader reliably better. The two caps share architecture, acquisition and inference; only the reader's training differs, and PC training cost $\approx 100\times$ more GPU time.

Zeros that are not measurements. Frozen ES/RET-ES are 0 **by the eligibility screen** (items the base already answered were excluded from every pool); frozen LS and near-miss preservation are 1 **by definition** (the frozen output is the reference). These cells are marked in every table and are never used to compute a relative improvement.

2. Which requested comparisons the existing data support

requested comparison	support	how	where
Basic: ES / RET-ES / RET-GS, all three models, three datasets	supported	readers: record checkpoints (zsRE/CounterFact) and this study's MQuAKE evaluations; frozen: greedy generations on the same prompts (zero by screen / measured zero)	benchmark_original.csv; cap_value/basic_performance.csv
Basic: teacher-forced target NLL, all three models	supported	scores.parquet (same serialised prompt/answer pairs, same code path)	cap_value/basic_performance.csv
Basic: locality vs frozen (direct) and vs own cap-off reference	supported	direct: teacher-forced NLL and decoded answers on the same locality/near-miss/unseen prompts; own reference: record LS / near-miss / false-fire metrics; cap-off = frozen for these readers	cap_value/collateral_vs_frozen.csv
Basic: MQuAKE multi-hop accuracy, all three models	supported	readers: registered composition endpoint (80 cases); frozen: cap-off exact match on the same post-edit questions (case-level any/all not recorded for the frozen path; its question accuracy is 0)	mquake_composition.csv
Ordinary adaptation plots and absolute/percentage improvements	supported	losses: percentage of frozen loss; success rates: percentage points only (frozen rate 0 makes relative change undefined)	cap_value/improvements.csv
Extremes A: each model's own worst 5 % / 1 % (CVaR95 / CVaR99) with CIs	supported	item-group bootstrap, 2,000 draws	cap_value/extremes_own_worst.csv
Extremes B: frozen-defined hardest 5 % / 1 %, all three models, with CIs	supported	fixed sets; bootstrap within the set	cap_value/extremes_frozen_hardest.csv
Actual loss by frozen-difficulty decile, all three models, CIs, shared axes	supported	500 group-bootstrap draws per decile	cap_value/deciles_actual_loss.csv; figures/cap_value/deciles_*.png
Tail shapes at common absolute thresholds, with n, sensitivity, uncertainty	supported (fits are finite-range; no power-law claim)	token-level pooled probe surprisal; frozen P90/P95/P97.5 and 5/8/10 nats; GPD only with >= 100 exceedances	cap_value/token_tails_common_thresholds.csv
Locality / unrelated text / collateral: each cap vs frozen and vs own cap-off, incl. severe events	supported	probes: direct per-probe deteriorations D with counts above 1/2/5 nats; unrelated text: record harm summaries (245,237 positions per cell)	cap_value/collateral_vs_frozen.csv; cap_value/harm_unrelated_text.csv
Negative findings: MQuAKE multi-hop; residual extreme losses when gating fails	supported	composition endpoint; abstained paraphrases keep the frozen loss exactly	cap_value/gating_failures.csv
Frozen multi-hop any-/all-question case success	partial	only question-level cap-off exact match is recorded (0 of 240 on every cell); case-level any/all for the frozen path are therefore 0 as well but are not separately stored	mquake_composition.csv (frozen_question_new_exact)
Frozen-model exact match on paraphrases of CounterFact/MQuAKE 'true' facts	supported (as loss)	greedy newline-stop decoding almost never completes a true answer; the true-target NLL is reported instead	benchmark_standardized.csv (edit/true)
New realizations / orders / training seeds	not available	would need new GPU evaluations outside the frozen record; not run (amendment: no unnecessary reruns)	—

No additional GPU evaluation was needed: every matched three-model comparison requested could be reconstructed from the recorded teacher-forced scores, generations, record checkpoints and harm vectors of

ext-20261009.

3. Denominator and probe-population audit

Finding: in the previous report the frozen rows at the **100-edit** horizon used denominators of 300 (all stream items) while the readers' 100-edit metrics use the first 100 items; and the paired/CVaR/decile tables at horizon 100 scored all 300 edit/paraphrase/locality probes against a 100-edit memory, so 200 of the 300 "taught" items were in fact not yet taught. Both are corrected: `aw/extremes/assemble.py` now restricts the frozen horizon-h rows to the first h stream items, and `aw/extremes/paired.py` (and this supplement) restrict item-level families to `stream_index` $\leq$ `h` (`at_horizon`). The main report was regenerated with the fix. Frozen rows still inconsistent after the fix: **0**; item-level families with probes outside the horizon in the corrected tables: **54** (counted on the raw scores, which deliberately keep all 300 items for both memories — the restriction is applied at analysis time).

dataset	horizon	frozen ES denominator	frozen RET-GS denominator (items)	reader ES denominators	reader RET-GS denominators
zsRE	100	100	100	100	100
zsRE	300	300	300	300	300
CounterFact	100	100	100	100	100
CounterFact	300	300	300	300	300
MQuAKE	100	100	100	100	100
MQuAKE	300	300	300	300	300

Probe populations (identical for every model): zsRE 1,350 probes (300 edit, 300 paraphrase, 300 item-locality, 50 locality, 100 + 100 near-miss, 50 + 50 revision, 100 unseen); CounterFact 2,300 (600 paraphrases, 900 item-locality capped at 3 per item); MQuAKE 1,890 (incl. 240 composition questions, scored separately under the registered endpoint, not through the stream memory). `tables/cap_value/denominator_audit_probes.csv` lists every (model, dataset, horizon, family) count.

4. Basic performance: all three models, all three datasets

zsRE (300 edits; 300 items; losses in nats/token)

model	ES	RET-ES	RET-GS	NLL taught prompt	NLL paraphrase	NLL original fact	LS (own ref.)	locality worse than frozen	near-miss	unseen false fires	multi-hop Q acc.	multi-hop all-3
Frozen GPT-2 (no CAP)	0.0 % ‡	0.0 %	0.0 % §	6.009	6.098	n/a	100.0 % ¶	0 % (reference)	100.0 %	0.0 %	n/a	n/a
BP-CAP (seed mean)	100.0 %	100.0 %	98.0 %	0.015	0.126	n/a	100.0 %	0.0 %	75.0 %	9.7 %	n/a	n/a
PC-CAP (seed mean)	100.0 %	100.0 %	96.6 %	0.015	0.203	n/a	100.0 %	0.0 %	87.0 %	3.7 %	n/a	n/a
BP-CAP seed 0	100.0 %	100.0 %	97.3 %	0.015	0.161	n/a	100.0 %	0.0 %	85.0 %	4.0 %	n/a	n/a
BP-CAP seed 1	100.0 %	100.0 %	99.0 %	0.015	0.077	n/a	100.0 %	0.0 %	67.0 %	14.0 %	n/a	n/a
BP-CAP seed 2	100.0 %	100.0 %	97.7 %	0.015	0.142	n/a	100.0 %	0.0 %	73.0 %	11.0 %	n/a	n/a
PC-CAP seed 0	100.0 %	100.0 %	94.7 %	0.015	0.320	n/a	100.0 %	0.0 %	91.0 %	2.0 %	n/a	n/a
PC-CAP seed 1	100.0 %	100.0 %	97.0 %	0.015	0.168	n/a	100.0 %	0.0 %	85.0 %	5.0 %	n/a	n/a
PC-CAP seed 2	100.0 %	100.0 %	98.0 %	0.015	0.120	n/a	100.0 %	0.0 %	85.0 %	4.0 %	n/a	n/a

CounterFact (300 edits; 300 items; losses in nats/token)

model	ES	RET-ES	RET-GS	NLL taught prompt	NLL paraphrase	NLL original fact	LS (own ref.)	locality worse than frozen	near-miss	unseen false fires	multi-hop Q acc.	multi-hop all-3
Frozen GPT-2 (no CAP)	0.0 % ‡	0.0 %	0.0 % §	6.956	7.369	6.382	100.0 % ¶	0 % (reference)	100.0 %	0.0 %	n/a	n/a
BP-CAP (seed mean)	99.7 %	99.7 %	81.5 %	0.046	1.312	5.222	100.0 %	0.0 %	99.7 %	0.3 %	n/a	n/a
PC-CAP (seed mean)	100.0 %	100.0 %	66.9 %	0.019	2.429	5.215	100.0 %	0.3 %	97.3 %	0.3 %	n/a	n/a
BP-CAP seed 0	100.0 %	100.0 %	81.0 %	0.019	1.362	5.215	100.0 %	0.0 %	100.0 %	0.0 %	n/a	n/a
BP-CAP seed 1	99.0 %	99.0 %	80.0 %	0.100	1.449	5.237	100.0 %	0.0 %	99.0 %	1.0 %	n/a	n/a
BP-CAP seed 2	100.0 %	100.0 %	83.5 %	0.019	1.125	5.215	100.0 %	0.0 %	100.0 %	0.0 %	n/a	n/a
PC-CAP seed 0	100.0 %	100.0 %	56.8 %	0.019	3.183	5.215	100.0 %	0.1 %	98.0 %	0.0 %	n/a	n/a
PC-CAP seed 1	100.0 %	100.0 %	77.8 %	0.019	1.598	5.215	100.0 %	0.3 %	95.0 %	1.0 %	n/a	n/a
PC-CAP seed 2	100.0 %	100.0 %	66.0 %	0.019	2.504	5.215	100.0 %	0.0 %	99.0 %	0.0 %	n/a	n/a

MQuAKE (300 edits; 300 items; losses in nats/token)

model	ES	RET-ES	RET-GS	NLL taught prompt	NLL para-phrase	NLL original fact	LS (own ref.)	locality worse than frozen	near-miss	unseen false fires	multi-hop Q acc.	multi-hop all-3
Frozen GPT-2 (no CAP)	0.0 % ‡	0.0 %	0.0 % §	5.171	6.048	4.579	100.0 % ¶	0 % (refer-ence)	100.0 %	0.0 %	0.0 %	—
BP-CAP (seed mean)	97.0 %	98.0 %	67.4 %	0.124	1.592	7.941	100.0 %	1.1 %	100.0 %	0.0 %	0.6 %	0.0 %
PC-CAP (seed mean)	100.0 %	100.0 %	59.6 %	0.015	2.087	8.010	100.0 %	1.1 %	100.0 %	0.0 %	0.4 %	0.0 %
BP-CAP seed 0	100.0 %	100.0 %	62.3 %	0.015	2.009	8.010	100.0 %	0.4 %	100.0 %	0.0 %	0.0 %	0.0 %
BP-CAP seed 1	91.0 %	94.0 %	72.3 %	0.342	1.276	7.804	100.0 %	0.5 %	100.0 %	0.0 %	0.8 %	0.0 %
BP-CAP seed 2	100.0 %	100.0 %	67.7 %	0.015	1.491	8.010	100.0 %	0.9 %	100.0 %	0.0 %	0.8 %	0.0 %
PC-CAP seed 0	100.0 %	100.0 %	60.0 %	0.015	2.099	8.010	100.0 %	0.5 %	100.0 %	0.0 %	0.0 %	0.0 %
PC-CAP seed 1	100.0 %	100.0 %	60.3 %	0.015	1.994	8.010	100.0 %	1.1 %	100.0 %	0.0 %	0.4 %	0.0 %
PC-CAP seed 2	100.0 %	100.0 %	58.3 %	0.015	2.168	8.010	100.0 %	0.7 %	100.0 %	0.0 %	0.8 %	0.0 %

‡ zero **by the eligibility screen**: every stream item was selected because frozen GPT-2 did not produce the answer (zsRE 0/10,720, CounterFact 0/20,391 screened; MQuAKE items pass the same screen). § measured zero (paraphrases were not part of the screen). ¶ 1 **by definition**: the locality reference *is* the frozen response. Frozen multi-hop question accuracy is the recorded cap-off exact match on the post-edit answer (0/240 on every cell); case-level all-3 success for the frozen path is not separately stored (it is 0 because no question is correct). Readers' ES/RET-ES/RET-GS/LS/near-miss/false-fire values are the registered record metrics; losses are this study's teacher-forced measurements on identical prompt/answer pairs.

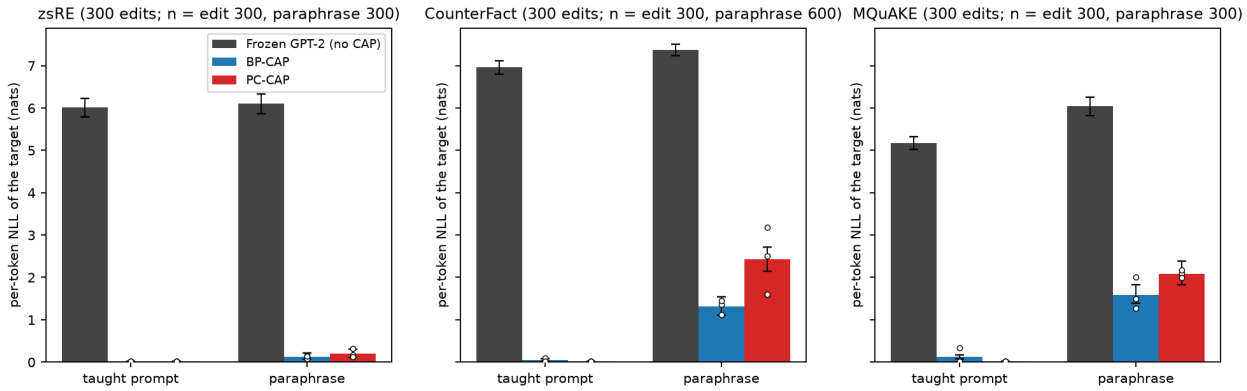


Figure 1: Ordinary adaptation: per-token loss

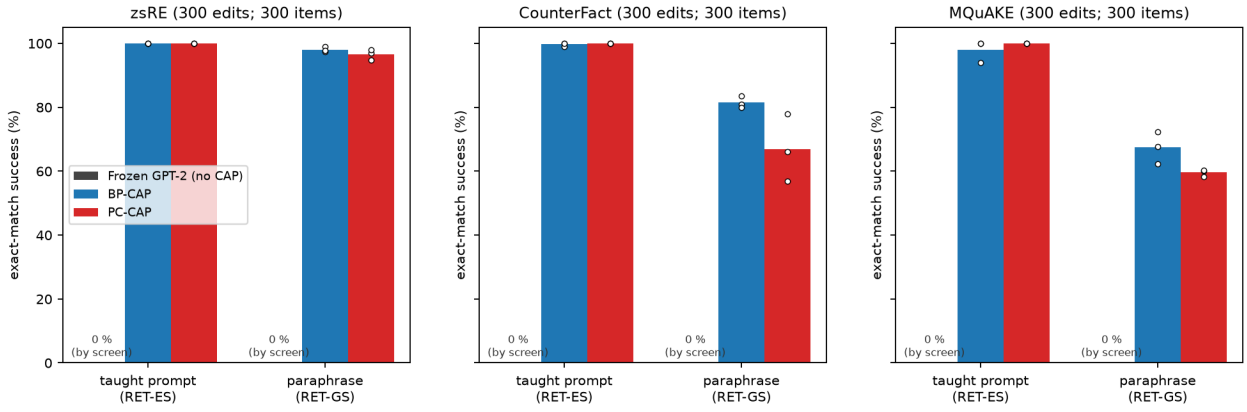


Figure 2: Ordinary adaptation: exact-match success

4.1 Absolute and percentage improvement over frozen GPT-2 (300 edits, seed means)

dataset	metric	frozen	BP-CAP	PC-CAP	BP-CAP vs frozen	PC-CAP vs frozen
zsRE	NLL taught prompt	6.009	0.015	0.015	−5.995 (99.8 %)	−5.995 (99.8 %)
zsRE	NLL paraphrase	6.098	0.126	0.203	−5.972 (97.9 %)	−5.895 (96.7 %)
zsRE	RET-ES	0.0 %	100.0 %	100.0 %	+100.0 points (rel. undefined, frozen 0)	+100.0 points (rel. undefined, frozen 0)
zsRE	RET-GS	0.0 %	98.0 %	96.6 %	+98.0 points (rel. undefined, frozen 0)	+96.6 points (rel. undefined, frozen 0)
CounterFact	NLL taught prompt	6.956	0.046	0.019	−6.911 (99.3 %)	−6.938 (99.7 %)
CounterFact	NLL paraphrase	7.369	1.312	2.429	−6.057 (82.2 %)	−4.941 (67.0 %)
CounterFact	NLL original fact	6.382	5.222	5.215	−1.160 (18.2 %)	−1.167 (18.3 %)
CounterFact	RET-ES	0.0 %	99.7 %	100.0 %	+99.7 points (rel. undefined, frozen 0)	+100.0 points (rel. undefined, frozen 0)
CounterFact	RET-GS	0.0 %	81.5 %	66.9 %	+81.5 points (rel. undefined, frozen 0)	+66.9 points (rel. undefined, frozen 0)
MQuAKE	NLL taught prompt	5.171	0.124	0.015	−5.047 (97.6 %)	−5.157 (99.7 %)
MQuAKE	NLL paraphrase	6.048	1.592	2.087	−4.456 (73.7 %)	−3.961 (65.5 %)
MQuAKE	NLL original fact	4.579	7.941	8.010	−-3.362 (-73.4 %)	−-3.430 (-74.9 %)
MQuAKE	RET-ES	0.0 %	98.0 %	100.0 %	+98.0 points (rel. undefined, frozen 0)	+100.0 points (rel. undefined, frozen 0)
MQuAKE	RET-GS	0.0 %	67.4 %	59.6 %	+67.4 points (rel. undefined, frozen 0)	+59.6 points (rel. undefined, frozen 0)
MQuAKE	multi-hop Q accuracy	0.0 %	0.6 %	0.4 %	+0.6 points (rel. undefined, frozen 0)	+0.4 points (rel. undefined, frozen 0)

Loss improvements are reported as absolute reductions and as a percentage of the frozen loss; success-rate improvements as percentage points, with the relative change marked undefined wherever the frozen rate is 0. Full per-seed and 100-edit rows: `tables/cap_value/improvements.csv`.

5. Extremes, comparison A: each model's own worst 5 % and 1 % (CVaR95 / CVaR99)

dataset	family	model	n	mean	own worst 5 % (CVaR95)	own worst 1 % (CVaR99)	max
zsRE	paraphrase	Frozen GPT-2 (no CAP)	300	6.098 [5.859, 6.332]	11.331 [10.773, 11.688]	12.161 [11.372, 12.588]	12.692
zsRE	paraphrase	BP-CAP (seed mean)	300	0.126 [0.059, 0.208]	2.046 [0.784, 3.523]	5.265 [2.559, 6.256]	6.433
zsRE	paraphrase	PC-CAP (seed mean)	300	0.203 [0.113, 0.309]	3.364 [1.817, 4.479]	5.750 [3.592, 6.332]	6.433
zsRE	edit	Frozen GPT-2 (no CAP)	300	6.009 [5.780, 6.227]	10.908 [10.214, 11.428]	11.870 [11.132, 12.150]	12.219
zsRE	edit	BP-CAP (seed mean)	300	0.015 [0.014, 0.016]	0.043 [0.038, 0.047]	0.052 [0.044, 0.058]	0.058
zsRE	edit	PC-CAP (seed mean)	300	0.015 [0.014, 0.016]	0.043 [0.038, 0.047]	0.052 [0.044, 0.058]	0.058
zsRE	unseen	Frozen GPT-2 (no CAP)	100	6.128 [5.743, 6.514]	10.353 [9.150, 11.353]	11.487 [9.717, 12.078]	12.078
zsRE	unseen	BP-CAP (seed mean)	100	6.240 [5.784, 6.687]	12.052 [10.184, 13.597]	13.617 [11.119, 15.035]	15.035
zsRE	unseen	PC-CAP (seed mean)	100	6.275 [5.844, 6.719]	11.838 [9.887, 13.434]	13.556 [10.894, 15.035]	15.035
zsRE	locality_item	Frozen GPT-2 (no CAP)	296	6.223 [6.012, 6.430]	10.333 [9.790, 10.810]	11.274 [10.717, 11.314]	11.314
zsRE	locality_item	BP-CAP (seed mean)	296	6.223 [6.012, 6.430]	10.333 [9.790, 10.810]	11.274 [10.717, 11.314]	11.314
zsRE	locality_item	PC-CAP (seed mean)	296	6.223 [6.012, 6.430]	10.333 [9.790, 10.810]	11.274 [10.717, 11.314]	11.314
CounterFact	paraphrase	Frozen GPT-2 (no CAP)	600	7.369 [7.230, 7.511]	10.396 [10.062, 10.736]	11.271 [10.581, 11.720]	11.931
CounterFact	paraphrase	BP-CAP (seed mean)	600	1.312 [1.095, 1.539]	7.929 [7.264, 8.422]	9.214 [8.515, 9.863]	10.381
CounterFact	paraphrase	PC-CAP (seed mean)	600	2.429 [2.148, 2.713]	8.784 [8.409, 9.050]	9.549 [9.063, 10.011]	10.381
CounterFact	edit	Frozen GPT-2 (no CAP)	300	6.956 [6.791, 7.110]	10.063 [9.522, 10.616]	11.097 [10.368, 11.635]	11.705
CounterFact	edit	BP-CAP (seed mean)	300	0.046 [0.019, 0.079]	0.554 [0.054, 1.173]	2.053 [0.058, 3.011]	3.162
CounterFact	edit	PC-CAP (seed mean)	300	0.019 [0.017, 0.020]	0.054 [0.051, 0.056]	0.058 [0.055, 0.059]	0.059
CounterFact	unseen	Frozen GPT-2 (no CAP)	100	6.924 [6.631, 7.205]	9.699 [9.202, 9.862]	9.866 [9.551, 10.008]	10.008
CounterFact	unseen	BP-CAP (seed mean)	100	6.923 [6.630, 7.203]	9.699 [9.202, 9.862]	9.866 [9.551, 10.008]	10.008
CounterFact	unseen	PC-CAP (seed mean)	100	6.923 [6.630, 7.203]	9.699 [9.202, 9.862]	9.866 [9.551, 10.008]	10.008
CounterFact	locality_item	Frozen GPT-2 (no CAP)	900	6.295 [6.150, 6.443]	10.026 [9.518, 10.414]	11.090 [10.565, 11.503]	12.008
CounterFact	locality_item	BP-CAP (seed mean)	900	6.289 [6.141, 6.438]	10.026 [9.518, 10.414]	11.090 [10.565, 11.503]	12.008
CounterFact	locality_item	PC-CAP (seed mean)	900	6.276 [6.129, 6.426]	10.026 [9.519, 10.414]	11.090 [10.565, 11.503]	12.008
MQuAKE	paraphrase	Frozen GPT-2 (no CAP)	300	6.048 [5.825, 6.260]	10.116 [9.469, 10.678]	11.374 [10.163, 12.015]	12.265
MQuAKE	paraphrase	BP-CAP (seed mean)	300	1.592 [1.382, 1.819]	6.840 [5.955, 7.748]	8.821 [7.061, 9.982]	10.446
MQuAKE	paraphrase	PC-CAP (seed mean)	300	2.087 [1.821, 2.378]	7.617 [7.010, 8.243]	8.924 [7.592, 10.016]	10.446
MQuAKE	edit	Frozen GPT-2 (no CAP)	300	5.171 [5.018, 5.326]	8.392 [7.797, 8.881]	9.425 [8.546, 9.831]	9.986
MQuAKE	edit	BP-CAP (seed mean)	300	0.124 [0.075, 0.176]	1.934 [1.147, 2.278]	2.665 [2.063, 2.800]	2.831
MQuAKE	edit	PC-CAP (seed mean)	300	0.015 [0.013, 0.016]	0.047 [0.041, 0.052]	0.055 [0.050, 0.059]	0.060
MQuAKE	unseen	Frozen GPT-2 (no CAP)	100	5.284 [5.012, 5.550]	7.985 [7.168, 9.019]	9.034 [7.394, 10.184]	10.184
MQuAKE	unseen	BP-CAP (seed mean)	100	5.284 [5.012, 5.550]	7.985 [7.168, 9.019]	9.034 [7.394, 10.184]	10.184
MQuAKE	unseen	PC-CAP (seed mean)	100	5.284 [5.012, 5.550]	7.985 [7.168, 9.019]	9.034 [7.394, 10.184]	10.184
MQuAKE	locality_item	Frozen GPT-2 (no CAP)	560	4.480 [4.323, 4.629]	7.717 [7.253, 8.157]	8.759 [8.034, 9.633]	10.456
MQuAKE	locality_item	BP-CAP (seed mean)	560	4.492 [4.337, 4.641]	7.762 [7.312, 8.186]	8.759 [8.077, 9.633]	10.456
MQuAKE	locality_item	PC-CAP (seed mean)	560	4.500 [4.342, 4.650]	7.917 [7.399, 8.471]	9.396 [8.161, 10.954]	11.951

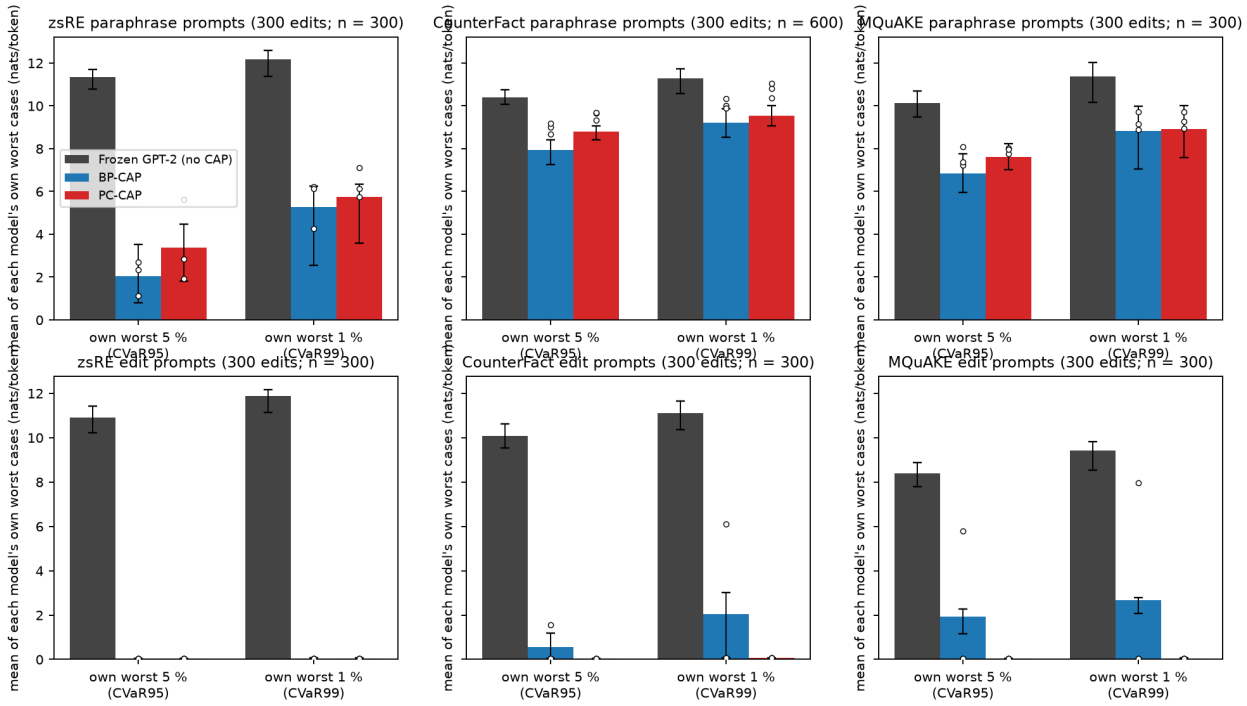


Figure 3: Own worst cases

6. Extremes, comparison B: the frozen model's hardest 5 % and 1 %, scored by all three models

The sets are fixed by the frozen per-token NLL (ties broken by probe id); every model is evaluated on exactly the same probes; intervals are bootstraps over the items inside the set.

dataset	family	fixed set	k	model	mean NLL on the set	exact-match on the set	share below 1 nat	max
zsRE	paraphrase	frozen-hardest 5 %	15	Frozen GPT-2 (no CAP)	11.365 [11.119, 11.656]	0.0 % [0.00, 0.00]	0.0 %	12.692
zsRE	paraphrase	frozen-hardest 5 %	15	BP-CAP (seed mean)	0.046 [0.020, 0.078]	100.0 % [1.00, 1.00]	100.0 %	0.214
zsRE	paraphrase	frozen-hardest 5 %	15	PC-CAP (seed mean)	0.046 [0.020, 0.078]	100.0 % [1.00, 1.00]	100.0 %	0.214
zsRE	paraphrase	frozen-hardest 1 %	3	Frozen GPT-2 (no CAP)	12.332 [12.030, 12.692]	0.0 % [0.00, 0.00]	0.0 %	12.692
zsRE	paraphrase	frozen-hardest 1 %	3	BP-CAP (seed mean)	0.045 [0.002, 0.124]	100.0 % [1.00, 1.00]	100.0 %	0.124
zsRE	paraphrase	frozen-hardest 1 %	3	PC-CAP (seed mean)	0.045 [0.002, 0.124]	100.0 % [1.00, 1.00]	100.0 %	0.124
zsRE	edit	frozen-hardest 5 %	15	Frozen GPT-2 (no CAP)	10.988 [10.659, 11.326]	0.0 % [0.00, 0.00]	0.0 %	12.219
zsRE	edit	frozen-hardest 5 %	15	BP-CAP (seed mean)	0.022 [0.013, 0.031]	100.0 % [1.00, 1.00]	100.0 %	0.058
zsRE	edit	frozen-hardest 5 %	15	PC-CAP (seed mean)	0.022 [0.013, 0.031]	100.0 % [1.00, 1.00]	100.0 %	0.058
zsRE	edit	frozen-hardest 1 %	3	Frozen GPT-2 (no CAP)	11.952 [11.693, 12.219]	0.0 % [0.00, 0.00]	0.0 %	12.219
zsRE	edit	frozen-hardest 1 %	3	BP-CAP (seed mean)	0.008 [0.003, 0.011]	100.0 % [1.00, 1.00]	100.0 %	0.011
zsRE	edit	frozen-hardest 1 %	3	PC-CAP (seed mean)	0.008 [0.003, 0.011]	100.0 % [1.00, 1.00]	100.0 %	0.011
CounterFact	paraphrase	frozen-hardest 5 %	30	Frozen GPT-2 (no CAP)	10.416 [10.213, 10.620]	0.0 % [0.00, 0.00]	0.0 %	11.931
CounterFact	paraphrase	frozen-hardest 5 %	30	BP-CAP (seed mean)	2.051 [0.891, 3.438]	78.9 % [0.64, 0.92]	66.7 %	10.381
CounterFact	paraphrase	frozen-hardest 5 %	30	PC-CAP (seed mean)	2.782 [1.495, 4.194]	74.4 % [0.60, 0.87]	56.7 %	10.381
CounterFact	paraphrase	frozen-hardest 1 %	6	Frozen GPT-2 (no CAP)	11.394 [11.206, 11.645]	0.0 % [0.00, 0.00]	0.0 %	11.931
CounterFact	paraphrase	frozen-hardest 1 %	6	BP-CAP (seed mean)	3.286 [0.253, 6.652]	55.6 % [0.22, 1.00]	50.0 %	9.987
CounterFact	paraphrase	frozen-hardest 1 %	6	PC-CAP (seed mean)	4.878 [1.007, 7.643]	55.6 % [0.27, 0.93]	33.3 %	9.106
CounterFact	edit	frozen-hardest 5 %	15	Frozen GPT-2 (no CAP)	10.126 [9.758, 10.519]	0.0 % [0.00, 0.00]	0.0 %	11.705
CounterFact	edit	frozen-hardest 5 %	15	BP-CAP (seed mean)	0.227 [0.013, 0.649]	97.8 % [0.93, 1.00]	93.3 %	3.162
CounterFact	edit	frozen-hardest 5 %	15	PC-CAP (seed mean)	0.018 [0.012, 0.025]	100.0 % [1.00, 1.00]	100.0 %	0.052
CounterFact	edit	frozen-hardest 1 %	3	Frozen GPT-2 (no CAP)	11.271 [10.680, 11.705]	0.0 % [0.00, 0.00]	0.0 %	11.705
CounterFact	edit	frozen-hardest 1 %	3	BP-CAP (seed mean)	0.008 [0.003, 0.014]	100.0 % [1.00, 1.00]	100.0 %	0.014
CounterFact	edit	frozen-hardest 1 %	3	PC-CAP (seed mean)	0.008 [0.003, 0.014]	100.0 % [1.00, 1.00]	100.0 %	0.014
MQuAKE	paraphrase	frozen-hardest 5 %	15	Frozen GPT-2 (no CAP)	10.170 [9.790, 10.631]	0.0 % [0.00, 0.00]	0.0 %	12.265
MQuAKE	paraphrase	frozen-hardest 5 %	15	BP-CAP (seed mean)	3.135 [2.228, 4.435]	33.3 % [0.13, 0.60]	0.0 %	10.446
MQuAKE	paraphrase	frozen-hardest 5 %	15	PC-CAP (seed mean)	3.135 [2.228, 4.435]	33.3 % [0.13, 0.60]	0.0 %	10.446
MQuAKE	paraphrase	frozen-hardest 1 %	3	Frozen GPT-2 (no CAP)	11.528 [11.055, 12.265]	0.0 % [0.00, 0.00]	0.0 %	12.265
MQuAKE	paraphrase	frozen-hardest 1 %	3	BP-CAP (seed mean)	4.301 [3.539, 5.812]	0.0 % [0.00, 0.00]	0.0 %	5.812
MQuAKE	paraphrase	frozen-hardest 1 %	3	PC-CAP (seed mean)	4.301 [3.539, 5.812]	0.0 % [0.00, 0.00]	0.0 %	5.812
MQuAKE	edit	frozen-hardest 5 %	15	Frozen GPT-2 (no CAP)	8.450 [8.131, 8.813]	0.0 % [0.00, 0.00]	0.0 %	9.986
MQuAKE	edit	frozen-	15	BP-CAP	0.566	93.3 %	80.0 %	2.831

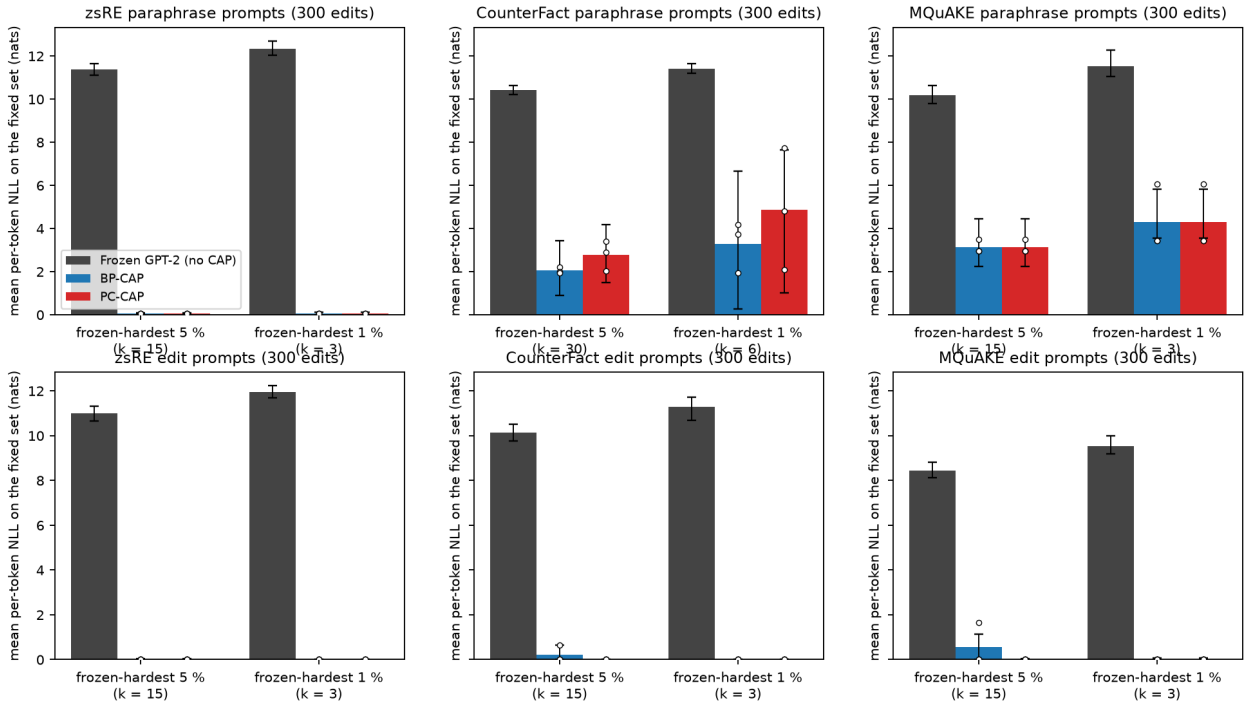


Figure 4: Frozen-hardest sets

6.1 Actual loss across frozen-difficulty deciles (not gains)

zsRE paraphrase prompts (300 edits) — mean NLL [95 % CI] / exact-match per decile

decile	n	frozen NLL range	Frozen GPT-2 (no CAP)	BP-CAP (seed mean)	PC-CAP (seed mean)
1	30	1.92–3.93	3.18 [2.93, 3.36] / 0 %	0.22 [0.01, 0.58] / 94 %	0.21 [0.02, 0.46] / 93 %
2	30	3.99–4.41	4.20 [4.15, 4.25] / 0 %	0.02 [0.02, 0.02] / 100 %	0.02 [0.02, 0.02] / 100 %
3	30	4.42–4.93	4.71 [4.65, 4.76] / 0 %	0.07 [0.01, 0.18] / 99 %	0.34 [0.02, 0.81] / 93 %
4	30	4.93–5.34	5.12 [5.07, 5.17] / 0 %	0.07 [0.01, 0.18] / 99 %	0.07 [0.01, 0.18] / 99 %
5	30	5.37–5.74	5.57 [5.53, 5.61] / 0 %	0.21 [0.02, 0.59] / 97 %	0.27 [0.02, 0.72] / 96 %
6	30	5.78–6.14	6.01 [5.97, 6.05] / 0 %	0.22 [0.02, 0.55] / 97 %	0.42 [0.08, 0.89] / 93 %
7	30	6.23–6.70	6.50 [6.45, 6.54] / 0 %	0.37 [0.01, 0.94] / 94 %	0.37 [0.01, 0.94] / 94 %
8	30	6.72–7.39	7.01 [6.94, 7.08] / 0 %	0.03 [0.02, 0.04] / 100 %	0.18 [0.02, 0.42] / 98 %
9	30	7.40–9.05	8.02 [7.85, 8.19] / 0 %	0.03 [0.02, 0.03] / 100 %	0.11 [0.02, 0.29] / 99 %
10	30	9.25–12.69	10.67 [10.38, 10.98] / 0 %	0.03 [0.02, 0.05] / 100 %	0.03 [0.02, 0.05] / 100 %

CounterFact paraphrase prompts (300 edits) — mean NLL [95 % CI] / exact-match per decile

decile	n	frozen NLL range	Frozen GPT-2 (no CAP)	BP-CAP (seed mean)	PC-CAP (seed mean)
1	60	3.62–5.51	4.90 [4.77, 5.01] / 0 %	0.96 [0.56, 1.39] / 79 %	1.68 [1.17, 2.17] / 64 %
2	60	5.53–6.08	5.82 [5.77, 5.86] / 0 %	0.86 [0.50, 1.32] / 86 %	1.46 [0.93, 2.01] / 75 %
3	60	6.08–6.44	6.27 [6.25, 6.30] / 0 %	0.75 [0.37, 1.14] / 88 %	1.95 [1.38, 2.48] / 69 %
4	60	6.45–6.86	6.68 [6.64, 6.71] / 0 %	0.98 [0.52, 1.53] / 86 %	2.22 [1.58, 2.86] / 67 %
5	60	6.86–7.38	7.10 [7.07, 7.14] / 0 %	1.38 [0.84, 1.99] / 81 %	2.61 [1.95, 3.36] / 64 %
6	60	7.39–7.79	7.60 [7.56, 7.63] / 0 %	1.26 [0.63, 1.90] / 82 %	1.82 [1.14, 2.53] / 75 %
7	60	7.80–8.15	7.97 [7.94, 8.00] / 0 %	1.75 [1.09, 2.45] / 77 %	2.96 [2.22, 3.70] / 62 %
8	60	8.16–8.66	8.37 [8.34, 8.41] / 0 %	1.72 [1.07, 2.40] / 79 %	3.00 [2.13, 3.92] / 65 %
9	60	8.70–9.38	8.99 [8.93, 9.04] / 0 %	2.04 [1.31, 2.80] / 74 %	3.81 [2.89, 4.75] / 56 %
10	60	9.39–11.93	10.00 [9.84, 10.15] / 0 %	1.42 [0.77, 2.23] / 83 %	2.78 [1.83, 3.63] / 71 %

MQuAKE paraphrase prompts (300 edits) — mean NLL [95 % CI] / exact-match per decile

decile	n	frozen NLL range	Frozen GPT-2 (no CAP)	BP-CAP (seed mean)	PC-CAP (seed mean)
1	30	1.44–3.83	2.92 [2.66, 3.14] / 0 %	0.47 [0.19, 0.82] / 86 %	0.72 [0.37, 1.16] / 76 %
2	30	3.84–4.27	4.06 [4.02, 4.11] / 0 %	0.96 [0.44, 1.56] / 77 %	1.40 [0.79, 2.05] / 67 %
3	30	4.29–4.94	4.57 [4.50, 4.66] / 0 %	1.27 [0.75, 1.92] / 71 %	1.88 [1.23, 2.64] / 60 %
4	30	4.97–5.59	5.25 [5.18, 5.32] / 0 %	1.83 [1.11, 2.62] / 62 %	2.01 [1.20, 2.91] / 59 %
5	30	5.59–6.06	5.82 [5.78, 5.87] / 0 %	2.00 [1.11, 2.93] / 67 %	2.00 [1.12, 2.97] / 67 %
6	30	6.07–6.54	6.33 [6.27, 6.38] / 0 %	1.62 [1.01, 2.16] / 73 %	3.00 [1.87, 3.98] / 52 %
7	30	6.55–6.99	6.74 [6.69, 6.79] / 0 %	1.42 [0.86, 2.09] / 67 %	2.30 [1.35, 3.33] / 53 %
8	30	6.99–7.56	7.28 [7.22, 7.34] / 0 %	1.94 [1.36, 2.66] / 56 %	2.76 [1.72, 3.79] / 49 %
9	30	7.64–8.56	8.02 [7.93, 8.11] / 0 %	1.58 [0.97, 2.34] / 80 %	1.95 [1.27, 2.77] / 77 %
10	30	8.57–12.27	9.49 [9.16, 9.83] / 0 %	2.83 [2.11, 3.68] / 37 %	2.85 [2.09, 3.74] / 37 %

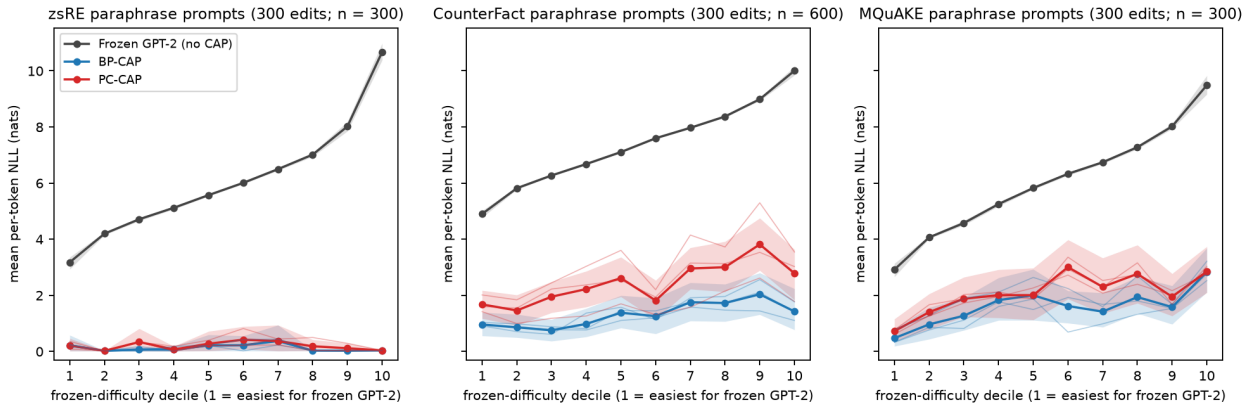


Figure 5: Deciles, paraphrase

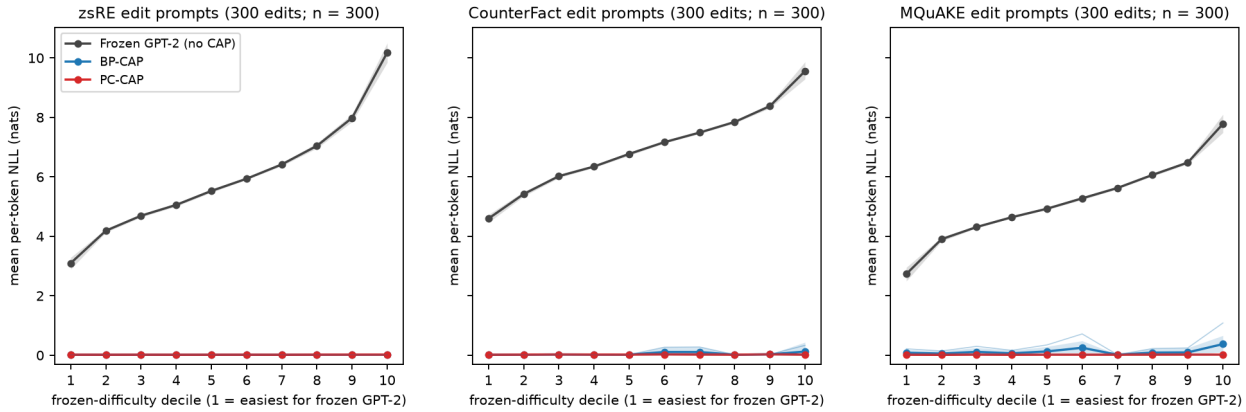


Figure 6: Deciles, taught prompt

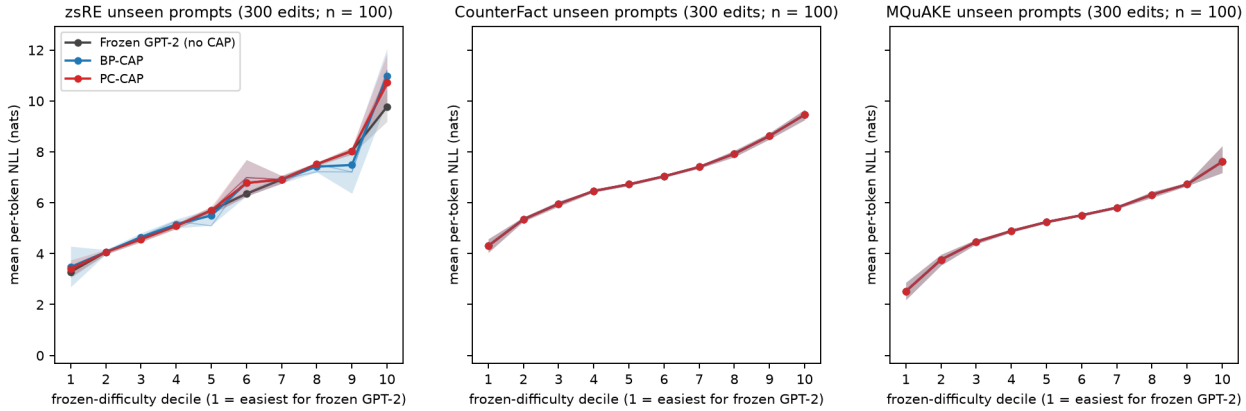


Figure 7: Deciles, un-taught prompt

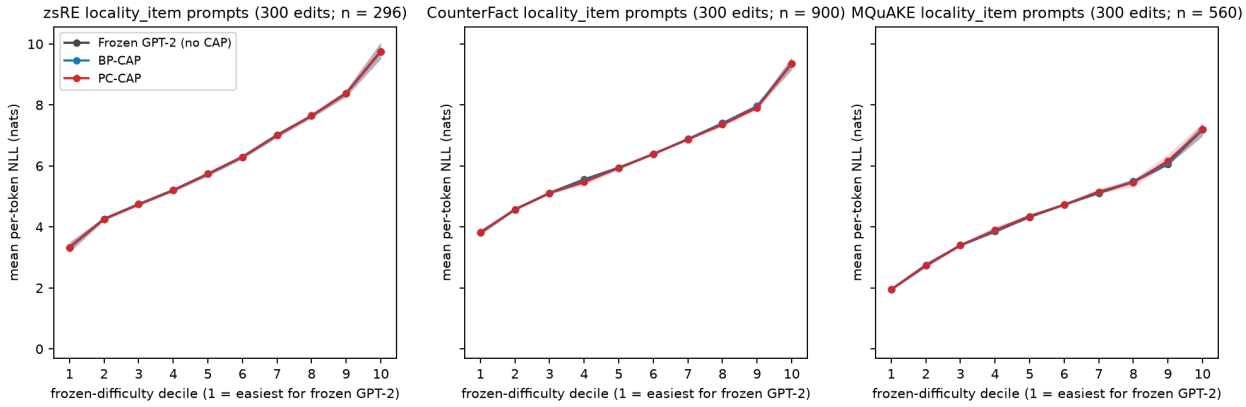


Figure 8: Deciles, item locality

7. Tail shape at common absolute thresholds

Variable: token-level target NLL (terminator excluded) pooled over the probe families where the cap should act (edit, paraphrase, unseen: new target) or abstain (item-locality, locality, near-miss neighbour: true target), at the 300-edit memory. Thresholds are **absolute and common to all models**: the frozen model's P90/P95/P97.5 and 5/8/10 nats. A generalized-Pareto shape is fitted only with ≥ 50 exceedances (flagged "exploratory" below 100), compared with the exponential special case on the same sample, with a 200-draw

item-group bootstrap interval. These are finite-range descriptive fits; a shape interval that includes or excludes zero says how the excess distribution decays above a given threshold, not that a power law holds.

7.1 Exceedance counts and shapes at the frozen P95, 8 nats and 10 nats

dataset	threshold	u (nats)	model	tokens	exceedances	fraction above u	mean excess	GPD shape κ [95 % CI]	log-lik gain/excess vs exponential
zsRE	frozen P95	14.01	Frozen GPT-2 (no CAP)	3,960	198	5.00 %	2.44	-0.42 [-0.53, -0.30]	0.0615
zsRE	frozen P95	14.01	BP-CAP seed 0	3,960	132	3.33 %	1.91	-0.19 [-0.37, -0.02]	0.0128
zsRE	frozen P95	14.01	BP-CAP seed 1	3,960	141	3.56 %	2.23	0.05 [-0.22, 0.17]	0.0020
zsRE	frozen P95	14.01	BP-CAP seed 2	3,960	137	3.46 %	2.13	-0.10 [-0.27, 0.03]	0.0041
zsRE	frozen P95	14.01	PC-CAP seed 0	3,960	125	3.16 %	1.95	-0.21 [-0.38, -0.03]	0.0157
zsRE	frozen P95	14.01	PC-CAP seed 1	3,960	136	3.43 %	2.07	-0.09 [-0.28, 0.04]	0.0038
zsRE	frozen P95	14.01	PC-CAP seed 2	3,960	133	3.36 %	1.99	-0.18 [-0.33, -0.04]	0.0114
zsRE	8 nats	8.00	Frozen GPT-2 (no CAP)	3,960	1373	34.67 %	3.25	-0.23 [-0.27, -0.18]	0.0215
zsRE	8 nats	8.00	BP-CAP seed 0	3,960	793	20.03 %	3.46	-0.30 [-0.36, -0.26]	0.0461
zsRE	8 nats	8.00	BP-CAP seed 1	3,960	795	20.08 %	3.59	-0.13 [-0.31, -0.10]	0.0181
zsRE	8 nats	8.00	BP-CAP seed 2	3,960	795	20.08 %	3.53	-0.24 [-0.31, -0.21]	0.0341
zsRE	8 nats	8.00	PC-CAP seed 0	3,960	794	20.05 %	3.41	-0.30 [-0.36, -0.26]	0.0443
zsRE	8 nats	8.00	PC-CAP seed 1	3,960	793	20.03 %	3.51	-0.25 [-0.32, -0.21]	0.0354
zsRE	8 nats	8.00	PC-CAP seed 2	3,960	790	19.95 %	3.48	-0.29 [-0.34, -0.25]	0.0428
zsRE	10 nats	10.00	Frozen GPT-2 (no CAP)	3,960	820	20.71 %	2.80	-0.19 [-0.25, -0.12]	0.0125
zsRE	10 nats	10.00	BP-CAP seed 0	3,960	507	12.80 %	2.84	-0.26 [-0.34, -0.20]	0.0320
zsRE	10 nats	10.00	BP-CAP seed 1	3,960	513	12.95 %	3.01	-0.09 [-0.28, -0.03]	0.0083
zsRE	10 nats	10.00	BP-CAP seed 2	3,960	511	12.90 %	2.94	-0.20 [-0.28, -0.15]	0.0208
zsRE	10 nats	10.00	PC-CAP seed 0	3,960	504	12.73 %	2.79	-0.25 [-0.32, -0.19]	0.0291
zsRE	10 nats	10.00	PC-CAP seed 1	3,960	507	12.80 %	2.92	-0.21 [-0.30, -0.15]	0.0227
zsRE	10 nats	10.00	PC-CAP seed 2	3,960	505	12.75 %	2.88	-0.25 [-0.31, -0.19]	0.0293
CounterFact	frozen P95	12.70	Frozen GPT-2 (no CAP)	2,150	108	5.02 %	1.34	-0.06 [-0.31, 0.20]	0.0012
CounterFact	frozen P95	12.70	BP-CAP seed 0	2,150	74	3.44 %	1.56	-0.07 [-0.35, 0.19]	0.0014
CounterFact	frozen P95	12.70	BP-CAP seed 1	2,150	74	3.44 %	1.46	-0.11 [-0.41, 0.19]	0.0035
CounterFact	frozen P95	12.70	BP-CAP seed 2	2,150	65	3.02 %	1.56	-0.10 [-0.34, 0.14]	0.0028
CounterFact	frozen P95	12.70	PC-CAP seed 0	2,150	72	3.35 %	1.51	-0.06 [-0.39, 0.28]	0.0011
CounterFact	frozen P95	12.70	PC-CAP seed 1	2,150	80	3.72 %	1.83	0.16 [-0.31, 0.40]	0.0203
CounterFact	frozen P95	12.70	PC-CAP seed 2	2,150	75	3.49 %	1.60	-0.20 [-0.49, 0.09]	0.0121
CounterFact	8 nats	8.00	Frozen GPT-2 (no CAP)	2,150	1018	47.35 %	2.30	-0.23 [-0.30, -0.20]	0.0308
CounterFact	8 nats	8.00	BP-CAP seed 0	2,150	601	27.95 %	2.35	-0.20 [-0.26, -0.15]	0.0199
CounterFact	8 nats	8.00	BP-CAP seed 1	2,150	609	28.33 %	2.34	-0.22 [-0.28, -0.18]	0.0246
CounterFact	8 nats	8.00	BP-CAP seed 2	2,150	582	27.07 %	2.29	-0.19 [-0.24, -0.15]	0.0193
CounterFact	8 nats	8.00	PC-CAP seed 0	2,150	681	31.67 %	2.31	-0.21 [-0.27, -0.17]	0.0254
CounterFact	8 nats	8.00	PC-CAP seed 1	2,150	606	28.19 %	2.44	-0.06 [-0.23, -0.01]	0.0044
CounterFact	8 nats	8.00	PC-CAP seed 2	2,150	651	30.28 %	2.34	-0.21 [-0.27, -0.16]	0.0224
CounterFact	10 nats	10.00	Frozen GPT-2 (no CAP)	2,150	492	22.88 %	1.75	-0.16 [-0.24, -0.09]	0.0119
CounterFact	10 nats	10.00	BP-CAP seed 0	2,150	289	13.44 %	1.88	-0.11 [-0.25, -0.02]	0.0056
CounterFact	10 nats	10.00	BP-CAP seed 1	2,150	294	13.67 %	1.85	-0.14 [-0.26, -0.04]	0.0087
CounterFact	10 nats	10.00	BP-CAP seed 2	2,150	274	12.74 %	1.80	-0.09 [-0.21, 0.03]	0.0033
CounterFact	10 nats	10.00	PC-CAP seed 0	2,150	328	15.26 %	1.77	-0.11 [-0.20, -0.04]	0.0058
CounterFact	10 nats	10.00	PC-CAP seed 1	2,150	301	14.00 %	1.98	0.02 [-0.21, 0.12]	0.0004
CounterFact	10 nats	10.00	PC-CAP seed 2	2,150	311	14.47 %	1.85	-0.12 [-0.24, -0.04]	0.0061
MQuAKE	frozen P95	11.60	Frozen GPT-2 (no CAP)	3,779	189	5.00 %	2.19	0.14 [-0.04, 0.28]	0.0056
MQuAKE	frozen P95	11.60	BP-CAP seed 0	3,779	75	1.98 %	2.12	0.30 [0.04, 0.51]	0.0269
MQuAKE	frozen P95	11.60	BP-CAP seed 1	3,779	58	1.53 %	1.53	0.17 [-0.24, 0.38]	0.0151
MQuAKE	frozen P95	11.60	BP-CAP seed 2	3,779	55	1.46 %	1.31	0.07 [-0.26, 0.26]	0.0016
MQuAKE	frozen P95	11.60	PC-CAP seed 0	3,779	67	1.77 %	2.07	0.32 [0.01, 0.66]	0.0309
MQuAKE	frozen P95	11.60	PC-CAP seed 1	3,779	68	1.80 %	1.67	0.20 [-0.17, 0.43]	0.0189
MQuAKE	frozen P95	11.60	PC-CAP seed 2	3,779	68	1.80 %	2.06	0.32 [0.04, 0.61]	0.0389
MQuAKE	8 nats	8.00	Frozen GPT-2 (no CAP)	3,779	874	23.13 %	2.45	-0.07 [-0.13, -0.02]	0.0032
MQuAKE	8 nats	8.00	BP-CAP seed 0	3,779	498	13.18 %	2.12	-0.03 [-0.11, 0.04]	0.0009
MQuAKE	8 nats	8.00	BP-CAP seed 1	3,779	480	12.70 %	1.94	-0.10 [-0.26, -0.04]	0.0085
MQuAKE	8 nats	8.00	BP-CAP seed 2	3,779	470	12.44 %	1.89	-0.18 [-0.27, -0.12]	0.0207
MQuAKE	8 nats	8.00	PC-CAP seed 0	3,779	497	13.15 %	2.05	-0.04 [-0.12, 0.03]	0.0012
MQuAKE	8 nats	8.00	PC-CAP seed 1	3,779	499	13.20 %	2.01	-0.08 [-0.23, -0.03]	0.0059
MQuAKE	8 nats	8.00	PC-CAP seed 2	3,779	502	13.28 %	2.05	-0.03 [-0.12, 0.03]	0.0007
MQuAKE	10 nats	10.00	Frozen	3,779	410	10.85 %	2.16	0.03 [-0.04,	0.0004

7.2 Threshold sensitivity (per-seed cells, three datasets)

threshold	model×dataset cells	fitted (≥ 50 exc.)	insufficient (< 50)	exploratory (50–99)	κ range	CI entirely > 0	CI entirely < 0	CI includes 0
frozen P90	21	21	0	0	-0.24 to 0.17	3	9	9
frozen P95	21	21	0	12	-0.42 to 0.32	3	4	14
frozen P97.5	21	7	14	7	-0.34 to 0.25	0	2	5
5 nats	21	21	0	0	-0.36 to -0.15	0	21	0
8 nats	21	21	0	0	-0.30 to -0.03	0	18	3
10 nats	21	21	0	0	-0.26 to 0.15	3	12	6

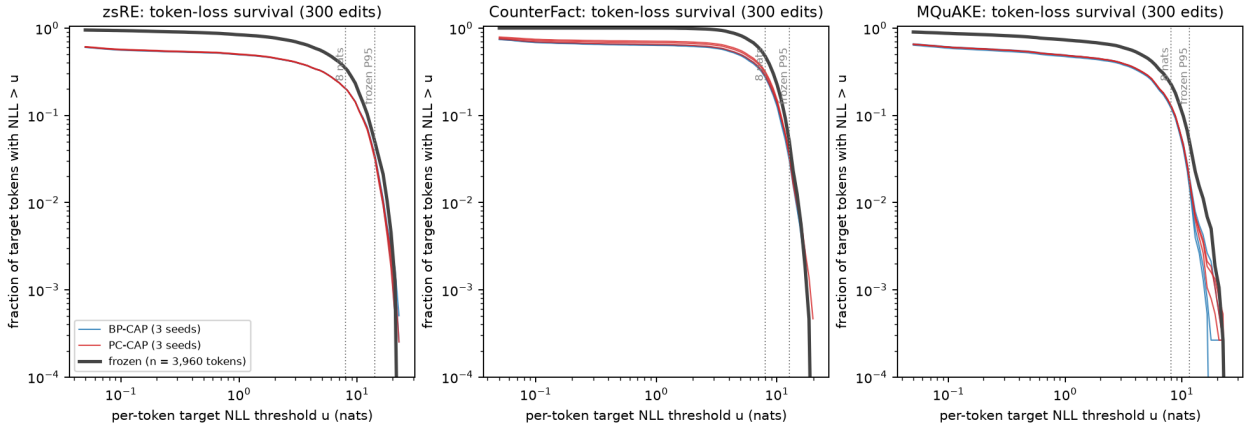


Figure 9: Token-loss survival

8. Locality, unrelated text and collateral damage: each cap vs the original frozen model and vs its own cap-off reference

For these GPT-2 small readers the ”own cap-off reference” of the record (the same base with the reader switched off) **is** frozen GPT-2: the direct forward reproduces the saved cap-off losses to 5e-5 nats. The two comparisons therefore coincide for unrelated text, and on probes the direct comparison below adds what the registered metrics cannot show: the size of each change, not only whether the decoded answer moved.

8.1 Per-probe deterioration D = L_cap − L_frozen on prompts the cap should leave alone (300 edits)

dataset	family	model	n	worse than frozen (> 1e-3 nats) [CI]	better than frozen	mean D	mean positive D	max D	severe D > 1 / 2 / 5 nats (3 seeds)	decoded answer changed vs frozen (seed mean)	record metric vs own cap-off (seeds 0–2)
zsRE	locality_item	BP-CAP (seed mean)	296	0.0 % [0.0, 0.0]	0.0 %	-0.0000	0.000	0.00	0 / 0 / 0	n/a (not generated)	—
zsRE	locality_item	PC-CAP (seed mean)	296	0.0 % [0.0, 0.0]	0.0 %	-0.0000	0.000	0.00	0 / 0 / 0	n/a (not generated)	—
zsRE	locality	BP-CAP (seed mean)	50	0.0 % [0.0, 0.0]	0.0 %	0.0000	0.000	0.00	0 / 0 / 0	0.0 %	LS = 100.0 %, 100.0 %, 100.0 %
zsRE	locality	PC-CAP (seed mean)	50	0.0 % [0.0, 0.0]	0.0 %	0.0000	0.000	0.00	0 / 0 / 0	0.0 %	LS = 100.0 %, 100.0 %, 100.0 %
zsRE	near_miss_neighbor	BP-CAP (seed mean)	100	14.0 % [7.0, 21.0]	7.0 %	0.1904	2.399	7.75	26 / 18 / 9	15.0 %	near_miss = 85.0 %, 67.0 %, 73.0 %
zsRE	near_miss_neighbor	PC-CAP (seed mean)	100	8.0 % [3.0, 13.0]	1.0 %	0.1262	1.992	7.75	11 / 8 / 4	6.0 %	near_miss = 91.0 %, 85.0 %, 85.0 %
zsRE	unseen	BP-CAP (seed mean)	100	10.0 % [5.0, 16.0]	4.0 %	0.1125	2.203	5.67	18 / 14 / 5	9.7 %	unseen_false_fire_rate = 4.0 %, 14.0 %, 11.0 %
zsRE	unseen	PC-CAP (seed mean)	100	5.0 % [1.0, 10.0]	0.0 %	0.1479	2.958	5.67	11 / 9 / 5	3.7 %	unseen_false_fire_rate = 2.0 %, 5.0 %, 4.0 %
CounterFact	locality_item	BP-CAP (seed mean)	900	0.0 % [0.0, 0.0]	0.7 %	-0.0063	0.000	0.00	0 / 0 / 0	n/a (not generated)	—
CounterFact	locality_item	PC-CAP (seed mean)	900	0.3 % [0.0, 0.8]	2.0 %	-0.0191	0.665	1.51	3 / 2 / 0	n/a (not generated)	—
CounterFact	locality	BP-CAP (seed mean)	50	0.0 % [0.0, 0.0]	0.0 %	-0.0000	0.000	0.00	0 / 0 / 0	0.0 %	LS = 100.0 %, 100.0 %, 100.0 %
CounterFact	locality	PC-CAP (seed mean)	50	0.0 % [0.0, 0.0]	0.0 %	-0.0000	0.000	0.00	0 / 0 / 0	0.0 %	LS = 100.0 %, 100.0 %, 100.0 %
CounterFact	near_miss_neighbor	BP-CAP (seed mean)	100	0.0 % [0.0, 0.0]	1.0 %	-0.0066	0.000	0.00	0 / 0 / 0	0.3 %	near_miss = 100.0 %, 99.0 %, 100.0 %
CounterFact	near_miss_neighbor	PC-CAP (seed mean)	100	1.0 % [0.0, 3.0]	4.0 %	-0.0478	0.030	0.03	0 / 0 / 0	2.7 %	near_miss = 98.0 %, 95.0 %, 99.0 %
CounterFact	unseen	BP-CAP (seed mean)	100	0.0 % [0.0, 0.0]	1.0 %	-0.0012	0.000	0.00	0 / 0 / 0	0.3 %	unseen_false_fire_rate = 0.0 %, 1.0 %, 0.0 %
CounterFact	unseen	PC-CAP (seed mean)	100	0.0 % [0.0, 0.0]	1.0 %	-0.0012	0.000	0.00	0 / 0 / 0	0.3 %	unseen_false_fire_rate = 0.0 %, 1.0 %, 0.0 %
MQuAKE	locality_item	BP-CAP (seed mean)	560	1.1 % [0.4, 2.0]	0.4 %	0.0123	1.313	1.92	10 / 4 / 1	n/a (not generated)	—
MQuAKE	locality_item	PC-CAP (seed mean)	560	1.1 % [0.4, 2.0]	0.4 %	0.0199	2.295	5.75	13 / 8 / 3	n/a (not generated)	—
MQuAKE	locality	BP-CAP (seed mean)	50	0.0 % [0.0, 0.0]	0.0 %	0.0000	0.000	0.00	0 / 0 / 0	0.0 %	LS = 100.0 %, 100.0 %, 100.0 %
MQuAKE	locality	PC-CAP (seed mean)	50	0.0 % [0.0, 0.0]	0.0 %	0.0000	0.000	0.00	0 / 0 / 0	0.0 %	LS = 100.0 %, 100.0 %, 100.0 %
MQuAKE	near_miss_neighbor	BP-CAP (seed mean)	100	0.0 % [0.0, 0.0]	0.0 %	-0.0000	0.000	0.00	0 / 0 / 0	0.0 %	near_miss = 100.0 %, 100.0 %, 100.0 %
MQuAKE	near_miss_neighbor	PC-CAP (seed mean)	100	0.0 % [0.0, 0.0]	0.0 %	-0.0000	0.000	0.00	0 / 0 / 0	0.0 %	near_miss = 100.0 %, 100.0 %, 100.0 %
MQuAKE	unseen	BP-CAP (seed mean)	100	0.0 % [0.0, 0.0]	0.0 %	0.0000	0.000	0.00	0 / 0 / 0	0.0 %	unseen_false_fire_rate = 0.0 %, 0.0 %, 0.0 %
MQuAKE	unseen	PC-CAP (seed mean)	100	0.0 % [0.0, 0.0]	0.0 %	0.0000	0.000	0.00	0 / 0 / 0	0.0 %	unseen_false_fire_rate = 0.0 %, 0.0 %, 0.0 %

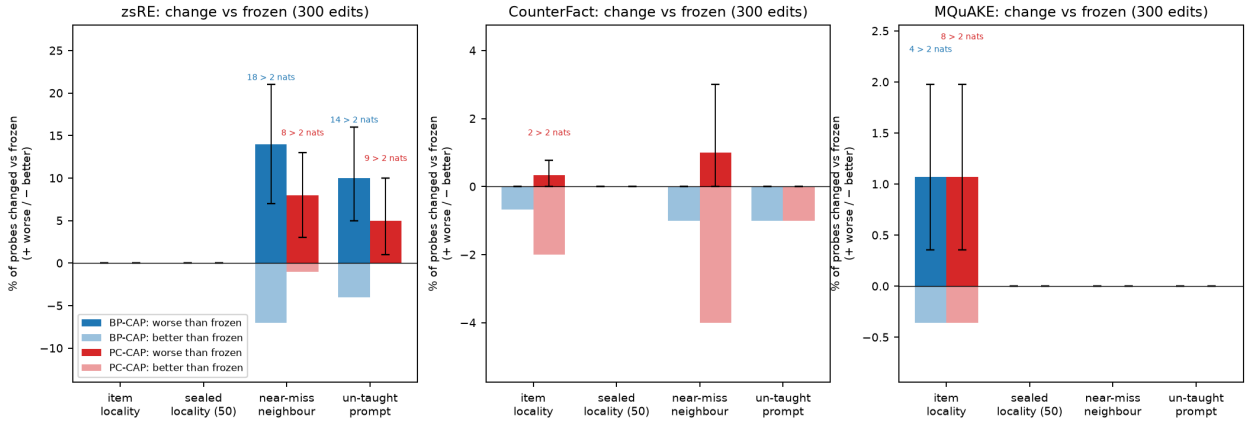


Figure 10: Collateral change vs frozen

8.1b Corrigendum (2026-10-20): item-locality prompts that are themselves taught items

40 of the item-locality prompts (all on MQuAKE; 40 rows, 39 distinct prompts of 600) are the exact prompts of other items taught in the same 300-edit stream, scored here against their *old* answers. A cap that fires on such a prompt is applying its own record for that item, and the registered new answer is the one it was taught; counting these rows as collateral damage overstated MQuAKE item-locality deterioration in the 10 October version of this report (7.7 % of prompts worse than frozen, about 33 severe events per seed). Since this date `aw/extremes/assemble.py` labels them `locality_item_taught`, the `locality_item` rows above are the genuinely unrelated stratum, and this table keeps the historical all-row figures beside both strata (found independently by Capex's 20 October audit and by Phase 0 of the post-conference programme). zsRE and CounterFact have no such overlap.

dataset	model	all rows (historical)	worse than frozen (historical)	severe D > 2 (historical)	rows that are taught items	fired on them	severe on them	genuinely unrelated rows	worse than frozen	severe	max D
CounterFact	BP-CAP seed 0	900	0.0 %	0	0	0	0	900	0.0 %	0	0.00
CounterFact	BP-CAP seed 1	900	0.0 %	0	0	0	0	900	0.0 %	0	0.00
CounterFact	BP-CAP seed 2	900	0.0 %	0	0	0	0	900	0.0 %	0	0.00
CounterFact	PC-CAP seed 0	900	0.1 %	1	0	0	0	900	0.1 %	1	2.26
CounterFact	PC-CAP seed 1	900	0.3 %	1	0	0	0	900	0.3 %	1	2.26
CounterFact	PC-CAP seed 2	900	0.0 %	0	0	0	0	900	0.0 %	0	0.00
MQuAKE	BP-CAP seed 0	600	7.0 %	31	40	40	31	560	0.4 %	0	1.39
MQuAKE	BP-CAP seed 1	600	7.2 %	32	40	40	31	560	0.5 %	1	5.75
MQuAKE	BP-CAP seed 2	600	7.5 %	34	40	40	31	560	0.9 %	3	3.68
MQuAKE	PC-CAP seed 0	600	7.2 %	32	40	40	31	560	0.5 %	1	5.75
MQuAKE	PC-CAP seed 1	600	7.7 %	35	40	40	31	560	1.1 %	4	5.75
MQuAKE	PC-CAP seed 2	600	7.3 %	34	40	40	31	560	0.7 %	3	5.75
zsRE	BP-CAP seed 0	296	0.0 %	0	0	0	0	296	0.0 %	0	0.00
zsRE	BP-CAP seed 1	296	0.0 %	0	0	0	0	296	0.0 %	0	0.00
zsRE	BP-CAP seed 2	296	0.0 %	0	0	0	0	296	0.0 %	0	0.00
zsRE	PC-CAP seed 0	296	0.0 %	0	0	0	0	296	0.0 %	0	0.00
zsRE	PC-CAP seed 1	296	0.0 %	0	0	0	0	296	0.0 %	0	0.00
zsRE	PC-CAP seed 2	296	0.0 %	0	0	0	0	296	0.0 %	0	0.00

8.2 Largest collateral deteriorations (two per family and dataset)

dataset	family	model	subject	L frozen	L cap	D	fired	rel. write
zsRE	near_miss_neighbour	BP-CAP seed 0	nan	6.84	14.59	7.75	True	0.121
zsRE	near_miss_neighbour	BP-CAP seed 1	nan	6.84	14.59	7.75	True	0.121
zsRE	unseen	PC-CAP seed 1	Stefan Glarner	6.53	12.88	6.35	True	0.151
zsRE	unseen	PC-CAP seed 2	Stefan Glarner	6.53	12.88	6.35	True	0.151
CounterFact	locality_item	PC-CAP seed 0	nan	6.15	8.41	2.26	True	0.140
CounterFact	locality_item	PC-CAP seed 1	nan	6.15	8.41	2.26	True	0.140
CounterFact	near_miss_neighbour	PC-CAP seed 1	nan	5.33	5.42	0.09	True	0.115
MQuAKE	locality_item	BP-CAP seed 1	nan	6.20	11.95	5.75	True	0.120
MQuAKE	locality_item	PC-CAP seed 2	nan	6.20	11.95	5.75	True	0.120

8.3 Unrelated text: 1,931 windows, 245,237 positions per cell; change = NLL(cap) minus NLL(frozen) at the same prefix [record]

dataset	cap	positions	mean Δ (nats)	ES99+ (positive part)	max Δ	positions Δ > 0.01	positions Δ > 1	fraction of positions changed
CounterFact	BP-CAP seed 0	245,237	2.02e-03	0.2070	8.52	227	149	1.04e-03
CounterFact	BP-CAP seed 1	245,237	5.35e-03	0.5513	12.50	567	363	2.69e-03
CounterFact	BP-CAP seed 2	245,237	1.18e-03	0.1208	9.86	138	83	6.08e-04
CounterFact	PC-CAP seed 0	245,237	1.28e-03	0.1324	7.71	162	112	7.54e-04
CounterFact	PC-CAP seed 1	245,237	6.28e-03	0.6413	12.36	636	419	2.94e-03
CounterFact	PC-CAP seed 2	245,237	1.95e-03	0.2061	12.62	291	155	1.44e-03
MQuAKE	BP-CAP seed 0	245,237	8.60e-04	0.0883	9.83	86	59	4.04e-04
MQuAKE	BP-CAP seed 1	245,237	5.00e-03	0.5129	13.14	635	357	2.96e-03
MQuAKE	BP-CAP seed 2	245,237	8.60e-04	0.0893	7.77	122	69	5.71e-04
MQuAKE	PC-CAP seed 0	245,237	1.09e-04	0.0116	4.53	19	10	9.79e-05
MQuAKE	PC-CAP seed 1	245,237	5.55e-03	0.5691	12.08	585	390	2.72e-03
MQuAKE	PC-CAP seed 2	245,237	1.15e-03	0.1159	11.19	105	76	4.61e-04
zsRE	BP-CAP seed 0	245,237	1.10e-04	0.0117	6.51	15	9	7.34e-05
zsRE	BP-CAP seed 1	245,237	1.25e-04	0.0127	7.92	12	9	5.30e-05
zsRE	BP-CAP seed 2	245,237	1.80e-04	0.0183	10.58	19	15	8.56e-05
zsRE	PC-CAP seed 0	245,237	0.00e+00	0.0000	0.00	0	0	0.00e+00
zsRE	PC-CAP seed 1	245,237	4.56e-05	0.0048	4.45	4	4	2.45e-05
zsRE	PC-CAP seed 2	245,237	0.00e+00	0.0000	0.00	0	0	0.00e+00

9. Negative findings, stated plainly

9.1 MQuAKE multi-hop composition

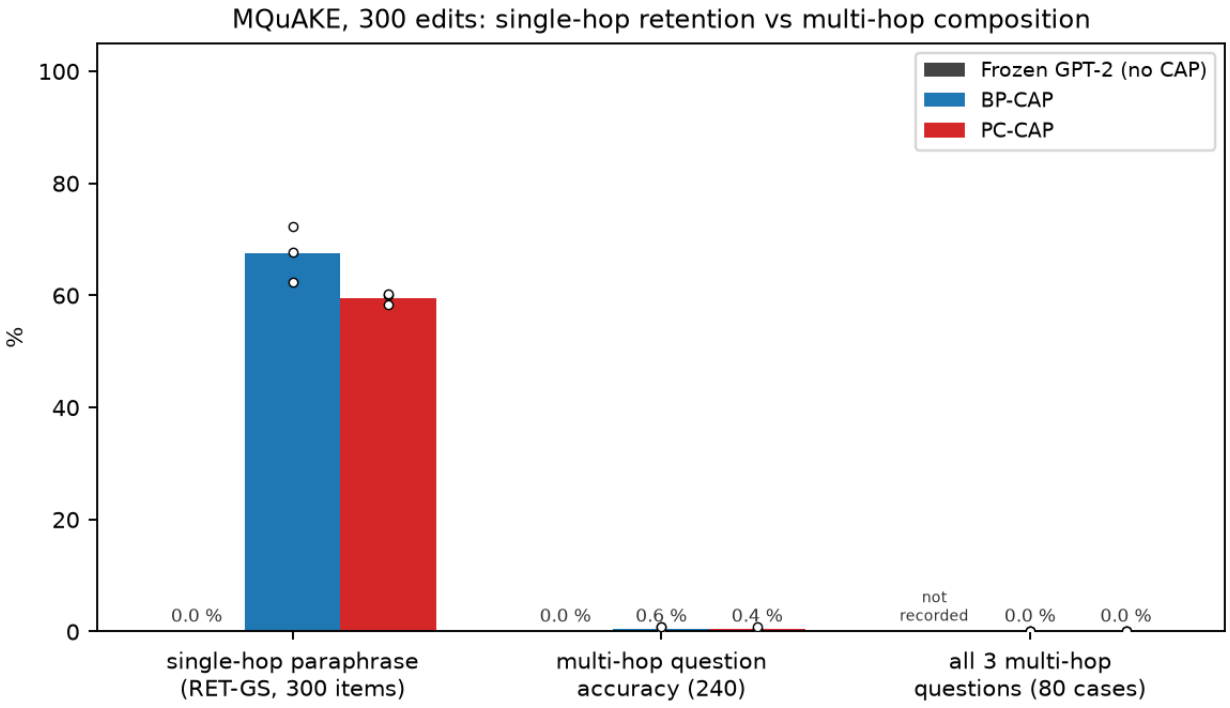


Figure 11: MQuAKE multi-hop

Both caps answer the single-hop paraphrases of the edited facts (RET-GS 58–72 %) and neither composes them: the registered composition endpoint (80 cases × 3 questions, dependency edits taught from a fresh state) gives all-three-questions success of 0/80 for every seed of both caps and question accuracy of 0–2 of 240; the frozen cap-off exact match on the same questions is 0/240. The Stage-4 selected v5 reader behaves the same (0–2 questions of 240–264 per cell). The CAP adds retrieval of single facts; it adds no multi-hop reasoning over them.

9.2 Residual extreme losses when gating fails

dataset	family	cap	n	abstained	NLL abstained (mean)	NLL abstained (CVaR95)	NLL fired (mean)	NLL fired (CVaR95)	success when fired	frozen-hardest-5 % abstained	share of own worst 5 % that abstained	fired but > 2 nats	max	L_abstained − L_frozen
zsRE	paraphrase	BP-CAP seed 0	300	2.3 %	5.36	6.43	0.037	0.40	99.7 %	0/15	46.7 %	1	0.000000	
zsRE	paraphrase	BP-CAP seed 1	300	0.7 %	6.08	6.43	0.036	0.40	99.7 %	0/15	13.3 %	1	0.000000	
zsRE	paraphrase	BP-CAP seed 2	300	2.0 %	5.29	6.43	0.037	0.40	99.7 %	0/15	40.0 %	1	0.000000	
zsRE	paraphrase	PC-CAP seed 0	300	5.3 %	5.63	7.93	0.021	0.09	100.0 %	0/15	100.0 %	0	0.000000	
zsRE	paraphrase	PC-CAP seed 1	300	2.7 %	4.97	6.43	0.037	0.40	99.7 %	0/15	53.3 %	1	0.000000	
zsRE	paraphrase	PC-CAP seed 2	300	2.0 %	4.98	6.43	0.021	0.09	100.0 %	0/15	40.0 %	0	0.000000	
zsRE	edit	BP-CAP seed 0	300	0.0 %	—	—	0.015	0.04	100.0 %	0/15	0.0 %	0	—	
zsRE	edit	BP-CAP seed 1	300	0.0 %	—	—	0.015	0.04	100.0 %	0/15	0.0 %	0	—	
zsRE	edit	BP-CAP seed 2	300	0.0 %	—	—	0.015	0.04	100.0 %	0/15	0.0 %	0	—	
zsRE	edit	PC-CAP seed 0	300	0.0 %	—	—	0.015	0.04	100.0 %	0/15	0.0 %	0	—	
zsRE	edit	PC-CAP seed 1	300	0.0 %	—	—	0.015	0.04	100.0 %	0/15	0.0 %	0	—	
zsRE	edit	PC-CAP seed 2	300	0.0 %	—	—	0.015	0.04	100.0 %	0/15	0.0 %	0	—	
CounterFact	paraphrase	BP-CAP seed 0	600	14.5 %	7.43	10.21	0.333	5.07	94.7 %	5/30	90.0 %	21	0.000004	
CounterFact	paraphrase	BP-CAP seed 1	600	16.0 %	7.58	10.55	0.281	4.12	95.2 %	5/30	100.0 %	18	0.000009	
CounterFact	paraphrase	BP-CAP seed 2	600	12.3 %	7.23	10.39	0.266	3.79	95.2 %	4/30	96.7 %	19	0.000004	
CounterFact	paraphrase	PC-CAP seed 0	600	41.2 %	7.42	10.07	0.222	3.04	96.6 %	8/30	100.0 %	9	0.000009	
CounterFact	paraphrase	PC-CAP seed 1	600	18.7 %	7.25	10.03	0.300	4.55	95.7 %	3/30	93.3 %	15	0.000009	
CounterFact	paraphrase	PC-CAP seed 2	600	31.7 %	7.37	10.73	0.249	3.69	96.6 %	9/30	100.0 %	11	0.000009	
CounterFact	edit	BP-CAP seed 0	300	0.0 %	—	—	0.019	0.05	100.0 %	0/15	0.0 %	0	—	
CounterFact	edit	BP-CAP seed 1	300	1.0 %	8.12	9.43	0.019	0.05	100.0 %	1/15	20.0 %	0	0.000000	
CounterFact	edit	BP-CAP seed 2	300	0.0 %	—	—	0.019	0.05	100.0 %	0/15	0.0 %	0	—	
CounterFact	edit	PC-CAP seed 0	300	0.0 %	—	—	0.019	0.05	100.0 %	0/15	0.0 %	0	—	
CounterFact	edit	PC-CAP seed 1	300	0.0 %	—	—	0.019	0.05	100.0 %	0/15	0.0 %	0	—	
CounterFact	edit	PC-CAP seed 2	300	0.0 %	—	—	0.019	0.05	100.0 %	0/15	0.0 %	0	—	
MQuAKE	paraphrase	BP-CAP seed 0	300	28.3 %	5.74	9.48	0.533	2.78	87.0 %	2/15	100.0 %	20	0.000000	
MQuAKE	paraphrase	BP-CAP seed 1	300	14.0 %	5.52	9.25	0.586	3.41	84.1 %	1/15	93.3 %	27	0.000000	
MQuAKE	paraphrase	BP-CAP seed 2	300	17.3 %	5.68	8.91	0.614	3.41	81.9 %	1/15	93.3 %	27	0.000000	
MQuAKE	paraphrase	PC-CAP seed 0	300	29.3 %	5.78	9.43	0.571	2.87	84.9 %	2/15	100.0 %	22	0.000000	
MQuAKE	paraphrase	PC-CAP seed 1	300	27.7 %	5.62	8.75	0.606	3.01	83.4 %	1/15	100.0 %	25	0.000000	
MQuAKE	paraphrase	PC-	300	30.7	5.64	8.83	0.634	3.58	84.1	1/15	93.3	23	0.000000	

Where the reader abstains (hard null), the cap's output is the frozen model's output and the loss is the frozen loss exactly (last column). The caps' remaining worst 5 % of paraphrase losses are therefore dominated by abstentions on frozen-hard prompts, plus a small number of fired-but-wrong cases above 2 nats/token.

10. Question B inside question A: PC-CAP vs BP-CAP

dataset	family	BP-CAP gain over frozen	PC-CAP gain over frozen	PC − BP (positive favours PC)	PC − BP per seed
zsRE	edit	5.995 [5.765, 6.211]	5.995 [5.765, 6.211]	0.000 [0.000, 0.000]	s0: 0.000 [0.000, 0.000]; s1: 0.000 [0.000, 0.000]; s2: 0.000 [0.000, 0.000]
zsRE	paraphrase	5.972 [5.708, 6.217]	5.895 [5.621, 6.146]	-0.076 [-0.136, -0.027]	s0: -0.159 [-0.282, -0.055]; s1: -0.092 [-0.176, -0.023]; s2: 0.021 [-0.048, 0.099]
CounterFact	edit	6.911 [6.750, 7.069]	6.938 [6.774, 7.091]	0.027 [0.000, 0.061]	s0: 0.000 [0.000, 0.000]; s1: 0.081 [0.000, 0.182]; s2: 0.000 [0.000, 0.000]
CounterFact	paraphrase	6.057 [5.807, 6.309]	4.941 [4.648, 5.228]	-1.117 [-1.313, -0.916]	s0: -1.821 [-2.168, -1.447]; s1: -0.149 [-0.357, 0.057]; s2: -1.379 [-1.666, -1.085]
MQuAKE	edit	5.047 [4.890, 5.199]	5.157 [5.004, 5.311]	0.109 [0.060, 0.162]	s0: 0.000 [0.000, 0.000]; s1: 0.328 [0.179, 0.485]; s2: 0.000 [0.000, 0.000]
MQuAKE	paraphrase	4.456 [4.193, 4.708]	3.961 [3.633, 4.261]	-0.495 [-0.651, -0.347]	s0: -0.090 [-0.299, 0.106]; s1: -0.718 [-0.943, -0.502]; s2: -0.677 [-0.882, -0.486]

dataset	cap	RET-GS seed mean (min–max)	near-miss	unseen false fires	multi-hop all-3	text harm ES99+ (seed mean)	text harm max Δ
zsRE	BP-CAP (seed mean)	98.0 % (97.3 %–99.0 %)	75.0 %	9.7 %	n/a	0.0143	10.58
zsRE	PC-CAP (seed mean)	96.6 % (94.7 %–98.0 %)	87.0 %	3.7 %	n/a	0.0016	4.45
CounterFact	BP-CAP (seed mean)	81.5 % (80.0 %–83.5 %)	99.7 %	0.3 %	n/a	0.2930	12.50
CounterFact	PC-CAP (seed mean)	66.9 % (56.8 %–77.8 %)	97.3 %	0.3 %	n/a	0.3266	12.62
MQuAKE	BP-CAP (seed mean)	67.4 % (62.3 %–72.3 %)	100.0 %	0.0 %	0.0 %	0.2302	13.14
MQuAKE	PC-CAP (seed mean)	59.6 % (58.3 %–60.3 %)	100.0 %	0.0 %	0.0 %	0.2322	12.08

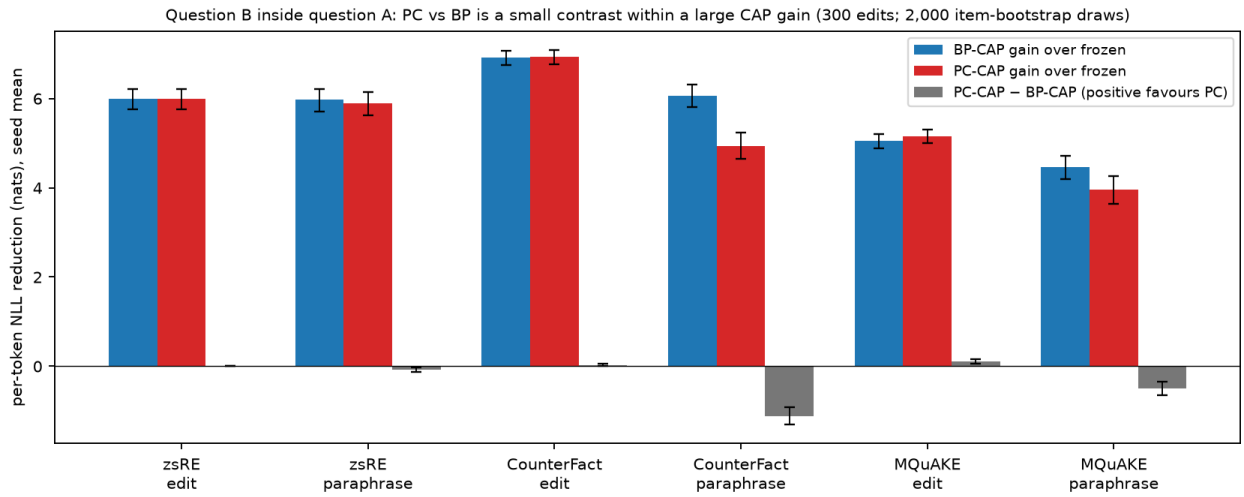


Figure 12: PC vs BP inside the CAP gain

11. Conclusions

A. Adding a CAP to frozen GPT-2 small converts the taught facts (own-prompt loss $\approx 6 \rightarrow \approx 0.02$ nats/token; retention 99–100 % vs 0 % by screen) and transfers to paraphrases the cap never saw (loss reductions of roughly 65–90 % of the frozen loss; retention 57–99 % vs a measured 0 %). The gain is at least as large on the frozen model's hardest decile and hardest 5 %/1 % as elsewhere, so the benefit is not confined to easy cases. Each cap's own worst 5 %/1 % are lower than frozen's but remain in the 5–10 nats/token range, because they are the cases where gating failed and the cap fell back to the frozen model. Against frozen GPT-2 directly, collateral change on locality-type prompts is rare and mostly tiny but includes a handful of severe events per seed, and on unrelated text the mean change per position is at most 6.3e-03 nats while single positions move by up to 13 nats (table 8.3). The CAP does not add multi-hop composition. Tail shapes at common thresholds are finite-range and do not change class; the caps thin the tail rather than reshape it.

B. Given that a CAP helps, the PC-trained reader is not better than the BP-trained one: the paired contrast is negative on all three datasets for paraphrases, PC-CAP retains fewer paraphrases on CounterFact and MQuAKE in every seed, and no extreme-case view (frozen-hardest sets, own worst cases, deciles, severe collateral events, text harm) shows a reliable PC advantage. PC training also costs $\approx 100\times$ the GPU time. The PC/BP difference is a difference of *gating* (which paraphrases trigger a write), small relative to the CAP-vs-frozen difference.

Qualifications. One subject realization and one teaching order; three training seeds on one population; GPT-2 small only; all intervals are within-population. Frozen exact-match rates on taught prompts are zero by construction; the frozen baseline is informative through teacher-forced losses, through paraphrase/locality decoded answers, and through its position in every three-model panel.

12. Reproducibility

- Code (CPU only): `aw/extremes/cap_value.py` (tables), `aw/extremes/cap_value_figures.py` (figures + slide package), `aw/extremes/cap_value_report.py` (this document); fixes in `aw/extremes/assemble.py` and `aw/extremes/paired.py` (horizon-restricted denominators). Pipeline: `results/extremes_analysis/ext-20261009/script`
  (now includes the three cap_value stages).
- Tables: `results/extremes_analysis/ext-20261009/tables/cap_value/` — basic_performance.csv, collateral_vs_frozen.csv, collateral_worst_cases.csv, deciles_actual_loss.csv, denominator_audit_probes.csv, denominator_audit_record.csv, extremes_frozen_hardest.csv, extremes_own_worst.csv, gating_failures.csv, harm_unrelated_text.csv, improvements.csv, support_matrix.csv, token_tails_common_thresholds.csv.
- Data: `assets/extremes_analysis/ext-20261009/data/cap_value/cases_seed_mean.parquet` (per-probe losses of all seven models and seed means), `survival_curves.parquet`.

- Figures: `assets/extremes_analysis/ext-20261009/figures/cap_value/` — adaptation_losses.png, adaptation_success.png, collateral_vs_frozen.png, deciles_edit.png, deciles_locality_item.png, deciles_paraphrase.png, deciles_unseen.png, frozen_hardest.png, mquake_multihop.png, own_worst.png, pc_vs_bp_within_gain.png, token_survival.png; slide package `figures/cap_value/slides/` — slide1_cap_value_ordinary.png, slide2_frozen_hard slide3_own_worst_and_tails.png, slide4_locality_and_limits.png, slide5_pc_vs_bp.png; captions in `figures/cap_value/captions.json`.
- Inputs: `scores.parquet`, `generations.parquet`, `paired_cases.parquet`, `benchmark_original.csv`, `mquake_composition.cs` `paired_gains.csv`, record checkpoints and `harm/summary.json` of the twelve PC-reader cells and the six MQuAKE cells (hashes in `manifest.json`).